Disruptive, Impulse-Control, & Conduct Disorders in South Korean Children & Adolescents

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Resumo (Abstract)

Objective Though disruptive, impulse-control, and conduct disorders (DICCD) represent significant public health challenges in Korea, diagnosis-specific epidemiological data remain limited. This study investigated DICCD's prevalence, risk factors, and psychiatric comorbidities as distinct diagnostic entities in a nationally representative sample of Korean children and adolescents. Methods Data from the National Mental Health Survey of Korea–Child & Adolescent 2022 (6,275 participants aged 6–17 years) were analyzed. Diagnoses were made using the Korean version of the Kiddie Schedule for Affective Disorders and Schizophrenia–Computerized Version, based on Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition. Sociodemographic and parental characteristics, lifestyle variables, and psychiatric comorbidities were assessed. Multivariable logistic regression analyses identified risk factors and comorbidity patterns. Results Prevalence rates were 2.76% for DICCD, 2.52% for oppositional defiant disorder (ODD), and 0.61% for conduct disorder (CD). ODD and CD shared risk factors—adolescent age (15–17 years), parental anxiety, and child sleep problems—but also exhibited distinct risk factor profiles. ODD was associated with parental depression, maternal employment, and excessive screen use, and CD with male sex and rural residence. Comorbidity patterns also differed. ODD was associated with mood disorders. CD exhibited elevated risks for substance use disorders and suicidal behavior. Conclusion This first nationally representative study demonstrated distinct prevalence rates and risk factors for DICCD, ODD, and CD in Korean children and adolescents. Disorder-specific comorbidity patterns require tailored prevention and treatment approaches. The findings underscore the clinical importance of maintaining diagnostic distinctions and implementing comprehensive psychiatric screening protocols for high-risk adolescents.

INTRODUCTION

Disruptive, impulse-control, and conduct disorders (DICCD) represent a significant public health challenge for children and adolescents worldwide [1]. Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5) restructured the classification of these disorders by consolidating oppositional defiant disorder (ODD) and conduct disorder (CD) from the former "Disruptive Behavior Disorders" category into the new eISSN 1976-3026 OPEN ACCESS "Disruptive, Impulse-Control, and Conduct Disorders" chapter, reflecting shared difficulties in emotional and behavioral self-control [2].

Within this framework, ODD is characterized by a persistent pattern of angry, argumentative, and vindictive behavior toward authority figures, whereas CD involves more severe violations of societal norms and the rights of others, including aggression, property destruction, deceitfulness, and serious rule violations [3]. Despite overlapping features, ODD and CD represent distinct diagnostic entities with different clinical presentations, developmental trajectories, and treatment requirements. Research has demonstrated that these disorders exhibit different patterns of psychiatric comorbidities, neurobiological correlates, and treatment responses, underscoring the clinical importance of maintaining diagnostic distinction.

Both disorders impose substantial personal, familial, and societal burdens, contributing to academic failure, interper-sonal difficulties, substance use, and antisocial behaviors that frequently persist into adulthood [4]. However, despite DSM-5's unified framework, epidemiological research has predominantly continued to examine these disorders either separately or in aggregate with other externalizing behaviors. Consequently, comprehensive epidemiological data investigating DICCD as a distinct diagnostic entity remain limited internationally, and are virtually absent in the Korean population.

In Korea, epidemiological data on DICCD remain fragmented and methodologically limited. Previous regional community studies of 7-to 12-year-old children reported an ODD prevalence of 3.7% [5], while international meta-analyses indicated that ODD prevalence ranged from 1% to 11% (mean, 3.3%); CD prevalence ranged from 1.9% to 2.1% [6 , 7]. However, these estimates were derived primarily from regional community samples, which significantly limit their generalizability to the broader Korean child and adolescent population. Critically, previous Korean research has frequently combined ODD and CD within the broader categories of externalizing behavioral problems or disruptive behavioral disorders, thereby obscuring the unique clinical features and epidemiological patterns of each condition. This methodological limitation prevents clinicians and researchers from understanding the distinct prevalence rates, risk factor profiles, and comorbidity patterns essential for developing targeted intervention strategies.

International research increasingly emphasizes the clinical importance of examining ODD and CD as separate diagnostic entities rather than as a combined category. Evidence suggests that although these disorders share some risk factors, they exhibit distinct patterns across the biological, psychological, and environmental domains. Prior research has identified multiple risk factors for these disorders, including parenting practices [8], parental psychopathology, peer relationship problems, parental unemployment, and various socioeconomic and sociodemographic factors such as sex [9 - 13]. Additionally, prenatal smoking and early childhood predictors, such as maternal depression, have been recognized as powerful predictors of later ODD or CD, offering critical opportunities for timely intervention before the onset of the full disorder [14 , 15]. Contemporary research has increasingly recognized the significance of modifiable family and environmental factors in the development of ODD and CD. However, the specific patterns and effects of sociodemographic risk factors on Korean children and adolescents remain unclear. This knowledge gap highlights the need for multidisciplinary approaches to prevention and intervention. Such an approach would facilitate the development of comprehensive evidence-based strategies that address the complex interplay among individual, familial, and environmental influences.

The high prevalence of psychiatric comorbidities further underscores the need for precise diagnosis and integrated care [16]. Research indicates that 92.4% of children and adolescents diagnosed with ODD experience at least one other psychiatric disorder during their lifetime, including anxiety, depression, impulse control, attention deficit/hyperactivity disorder (ADHD), and substance use disorders (SUDs) [5 , 17 , 18]. While ADHD comorbidity in ODD and CD is well-documented in prior research [5 , 6 , 17 , 19], the broader spectrum of neurodevelopmental disorder associations and their differential patterns across ODD and CD remain less comprehensively characterized. Moreover, the specific comorbidity profiles of psychiatric conditions in patients with ODD and CD and their clinical significance in different populations require further investigation. Understanding the comorbidity profiles of ODD and CD is critical for comprehensive assessment, collaborative care across specialties, and implementation of integrated treatment approaches that effectively address co-occurring conditions.

This study aimed to provide the first nationally representative diagnosis-specific epidemiological data on DICCD, ODD, and CD in Korean children and adolescents. By treating these disorders as distinct entities rather than as aggregated categories, this study will elucidate differences in prevalence rates, sociodemographic risk factors, early predictors, and comorbidity patterns essential for precise clinical assessment and targeted intervention strategies. Understanding the distinct epidemiological profiles of DICCD, ODD, and CD will enable Korean mental health professionals to develop more nuanced diagnostic approaches, implement early identification protocols, and design integrated multidisciplinary intervention programs tailored to the needs of children and adolescents with these challenging conditions. Based on prior literature [5 - 15 , 17 - 19], we hypothesized that: 1) ODD and CD would demonstrate distinct prevalence rates and sociodemographic correlates in Korean children and adolescents [5 - 7]; 2) both disorders would share certain risk factors (e.g., parental psychopathology) but also exhibit disorder-specific risk profiles across demographic, familial, and environmental domains [8 - 15]; and 3) ODD and CD would demonstrate differential psychiatric comorbidity patterns [17 - 19].

METHODS

Sample

We analyzed data from the National Mental Health Survey of Korea–Child & Adolescent 2022 (NMHSK-CA 2022) conducted between September 2022 and February 2023 by the National Center for Mental Health, Ministry of Health and Welfare, South Korea. The NMHSK-CA 2022 survey measured the lifetime and current prevalence of major psychiatric disorders and their correlation with sociodemographic characteristics and comorbidities among children and adolescents per DSM-5. This survey targeted a nationally representative sample of 6,275 children and adolescents aged 6 to 17 years, residing in general households across South Korea.

A stratified cluster sampling method was employed using census tracts and households as the primary and secondary sampling units based on the 2020 Korea Census. Trained interviewers administered the nationwide survey during structured home visits. All interviewers had prior experience with mental health surveys and had undergone standardized training, which included theoretical instruction, practical exercises, and evaluation. Data were collected through tablet-based structured interviews and questionnaires. Parents and adolescents completed the Korean version of the Kiddie Schedule for Affective Disorders and Schizophrenia–Computerized Version (KSADS-COMP), and additional instruments were used to assess sociodemographic and behavioral factors. The overall response rate was 38.8%, which increased to 59.9% after excluding non-contactable and ineligible households. Sampling weights were computed to account for design effects, non-response adjustments, and post-stratification based on national census distributions [20].

This study was approved by the Institutional Review Board of the National Center for Mental Health (IRB No. 116271-2025-10).

Measurement

Psychiatric disorders

DICCD, ODD, CD, and other psychiatric disorders (mood, anxiety, obsessive-compulsive, neurodevelopmental, substance use, eating, and trauma-related disorders) based on DSM-5 were

assessed using the Korean version of the KSADS-COMP, a web-based structured diagnostic interview aligned with DSM-5 criteria. The KSADS-COMP was translated and adapted for the Korean population; its validity has been established through two comprehensive validation studies [21 , 22]. The instrument comprises three stages: demographic information, screening questions, and diagnostic modules, accessible only through a web-based platform (http://ksads-comp.com). Caregivers (parents) completed the assessment for children aged 6–11 years, whereas adolescents aged 12–17 years responded directly. The lifetime diagnoses were established by evaluating a participant's historical and current symptoms against DSM-5 diagnostic criteria. For the participants who met specific symptoms counts and durations (e.g., at least 6 months for ODD and 12 months for CD), current diagnosis is defined as meeting the diagnostic criteria based on symptom frequency over the past 2 weeks, with symptoms rated on a 5-point scale (not at all, rarely, several days, more than half the days, and nearly every day) [23]. Only current diagnoses were analyzed in this study.

Sociodemographic and parental characteristics

Sociodemographic and parental variables were collected using standardized structured caregiver interviews. Variables included age group (6–8, 9–11, 12–14, 15–17), sex, residential area (urban vs. rural), parental marital status, parental employment status, household income (<500 ten thousand Won vs. ≥500 ten thousand Won), and number of children in the household. In addition, parental mental health and substance use were assessed as family-related demographic characteristics.

Parental depression and anxiety symptoms were evaluated using validated Korean versions of standardized self-report screening instruments. Each item was rated on a 4-point scale (0=not at all, 3=nearly every day). Depression was measured using the Patient Health Questionnaire-9 [24], which comprises nine items assessing depressive symptoms over the past 2 weeks. Total scores range from 0 to 27, and scores ≥9 are considered clinically significant. Anxiety symptoms were assessed using the Korean version of the Generalized Anxiety Disorder-7 scale [25], which consists of seven items assessing symptoms over the past 2 weeks. Total scores range from 0 to 21, with scores ≥5 classified as clinically significant.

Parental substance use was assessed using standardized measures. Alcohol use was evaluated using the Korean version of the Alcohol Use Disorders Identification Test [26]. The assessment included the frequency of alcohol consumption, categorized as no drinking, once a month or less, 2–4 times a month, and 2 times a week or more. Parental smoking behavior was measured using the Korean version of the Fagerstrom Test for Nicotine Dependence [27]. Current smoking status was recorded as a binary variable (yes/no).

Lifestyle

Physical activity was measured using a single item adapted from the Ontario Child Health Study [28] and translated into Korean. Participants or their caregivers reported the number of days during the past week in which the participant engaged in ≥60 minutes of moderate-to-vigorous physical activity. Additional examples of physical activities were adapted from Kim et al. [29] and translated into Korean. Responses were categorized into three groups (0 days, 1–2 days, and ≥3 days per week).

Sleep patterns were assessed using items adapted from Kim's study [29], translated, and modified for the Korean population. The assessment included three main domains (sleep delay, sleep interruption, and total sleep duration) in the past 6 months. Participants were classified as having sleep problems if they met at least one of the following criteria: 1) taking more than 31 minutes to fall asleep, 2) waking up in the middle of the night at least once a week, or 3) having a total sleep

duration of less than 5 hours or more than 10 hours per night. Sleep problems were recorded using binary response options (yes/no). They were intended to capture markedly short or long sleep duration and prominent sleep disturbances rather than fine-grained, age-specific variation in sleep patterns.

Screen time was measured using items adapted from Kim's study [29] and translated into Korean with response categories modified for this study. Participants or their caregivers were asked to report the average daily time spent on screen media in the past week. Four domains of screen time were assessed using a questionnaire: video use time (daily hours spent watching videos for enjoyment, including TV shows, movies, or YouTube), social media use time (daily hours spent on social media applications [Facebook, Instagram, KakaoTalk]), game use time (daily hours spent sitting and playing games [both online and offline]), and educational screen time (daily hours spent using electronic devices [computer, tablet, smartphone] for educational purposes, such as schoolwork or online learning). Each question asked respondents to rate their average daily usage time for the past week using a 5-point scale consisting of "less than 1 hour," "1–3 hours," "3–5 hours," "5–7 hours," and "7 hours or more" per day.

Game addiction

Game addiction was assessed by using an age-specific instrument. For children aged 6–11 years, caregivers completed the Korean Internet Addiction Observer Scale for Children and Adolescents (K-scale-based), developed by the National Information Society Agency [30]. For adolescents aged 12–17 years, the Structured Clinical Interview for Internet Gaming Disorder31 was used in accordance with the DSM-5 criteria.

Suicide risk indicators

Suicide risk included three indicators. Non-suicidal selfinjury, suicidal ideation, and suicidal behavior were assessed using the suicide-related module (L. Suicide) of the KSADSCOMP. The items were administered directly to adolescents aged 12–17 years or caregivers of children aged 6–11 years, and responses were recorded using binary response options (yes/no).

Statistical analysis

Group differences between participants with and without DICCD, ODD, and CD were assessed using the chi-square test with Rao-Scott corrections. Multivariable logistic regression analyses were performed to estimate adjusted odds ratios (ORs) and 95% confidence intervals (CIs) for the association between sociodemographic, parental, and lifestyle variables and the current prevalence of DICCD, ODD, and CD. Covariates for the multivariable models were selected a priori on the basis of theoretical relevance established in prior literature on disruptive behavior disorders. Specifically, sex, age group, maternal employment status, parental depression, and parental anxiety were treated as core confounders, as these variables have been consistently identified as established risk factors for ODD and CD [8 - 15]. To examine comorbidity patterns, additional logistic regression models tested the association of DICCD, ODD, and CD with other psychiatric disorders, game addiction, and suicidality. ORs and 95% CIs were calculated both with and without adjustment for the same a priori covariates. Because the analyses were guided by a priori hypotheses derived from established literature rather than by exploratory data mining, formal correction for multiple comparisons was not applied; this decision was also informed by the rarity of CD in the sample, where overly conservative corrections may obscure clinically meaningful associations. All analyses were conducted in SAS (version 9.4, SAS Institute Inc.), with statistical significance set at $p<0.05$; given the large number of

tests, findings—particularly those with p values approaching 0.05— should be interpreted with caution.

RESULTS

Prevalence, correlates, and comorbidities of DICCD

The estimated prevalence of DICCD was 2.76% (173/6,275). Participants with DICCD were significantly more likely to be male (p=0.015), aged 12–17 years (p<0.001), and have employed mothers (p=0.001) (Table 1). Parental depression (p< 0.001), parental anxiety (p<0.001), and child sleep problems (p=0.003) were significantly associated with DICCD. Maternal drinking (≥2–4 times a month) (p=0.016) also showed significance. Excessive screen use was related to DICCD, with higher video use (≥3 hours daily) (p=0.001), game use (≥1 hour daily) (p=0.001), and educational screen time (≥3 hours daily) (p<0.001) compared to non-DICCD participants (Table 2).

After adjusting for sex, age, maternal employment status, parental depression, and parental anxiety problems, mood disorders (OR 34.78, 95% CI 12.31–98.28), SUD (OR 10.46, 95% CI 5.60–19.52), neurodevelopmental disorder (OR 9.23, 95% CI 4.50–18.91), and suicidal behavior (OR 6.95, 95% CI 2.59–18.69) were significantly related to DICCD (Table 3).

Prevalence, correlates, and comorbidities of ODD

The estimated ODD prevalence was 2.52% (158/6,275). Participants with ODD were significantly more likely to be male (p=0.029) and aged 15–17 years (p<0.001) and have higher maternal employment rates (p<0.001), parental depression (p<0.001), and parental anxiety problems (p<0.001). Maternal drinking (2–4 times a month) (p=0.025) was also significant (Table 1). Related to screen time behaviors, participants with ODD had excessive video use (≥3 hours daily) (p=0.001), had game use (≥1 hour daily) (p=0.001), and had educational screen time (≥3 hours daily) (p<0.001) compared to those without ODD. Sleep problems was also associated with ODD (p=0.014) (Table 2).

After adjusting for sex, age, maternal employment status, parental depression, and parental anxiety problems, mood disorders (OR 37.43, 95% CI 13.27–105.59), neurodevelopmental disorders (OR 8.23, 95% CI 3.87–17.50), SUD (OR 5.86, 95% CI 2.81–12.22), and suicidal behavior (OR 5.57, 95% CI 2.15–14.44) were significantly related to ODD (Table 3).

Prevalence, correlates, and comorbidities of CD

The estimated prevalence of CD was 0.61% (38/6,275). Participants with CD were significantly more likely to be male (p=0.024) and aged 9–17 years (p<0.001), live in rural areas (p=0.039), and have parental anxiety problems (p=0.035) (Table 1). In addition, sleep problems (p=0.035) and moderate game use (1–3 hours daily) (p=0.028) were associated with a higher CD rate (Table 2).

After adjusting for sex, age, maternal employment status, parental depression, and parental anxiety problems, SUD (OR 60.49, 95% CI 24.59–148.81), suicidal behavior (OR 16.93, 95% CI 4.61–62.15), suicidal ideation (OR 15.19, 95% CI 5.51–41.92), mood disorders (OR 13.06, 95% CI 3.75–45.51), NSSI (OR 9.30, 95% CI 3.26–26.56), stress related disorders (OR 9.11, 95% CI 1.44–57.54), and anxiety disorders (OR 8.14, 95% CI 1.59–41.67) were significantly related to CD (Table 3).

DISCUSSION

This study was the first comprehensive, nationally representative epidemiological investigation of DICCD in Korean children and adolescents. Specifically, this study investigated the prevalence, risk factors, and comorbidities of a wide range of psychiatric disorders, including DICCD, ODD, and CD, using a nationally representative sample.

This study found the estimated prevalence of DICCD to be 2.76%. DSM-5 significantly reorganized behavioral disorder classifications: ADHD was moved to neurodevelopmental disorders, while the new “Disruptive, Impulse-Control, and Conduct Disorders” category now encompasses ODD, CD, intermittent explosive disorder, kleptomania, and pyromania. To the best of our knowledge, this is the first nationwide epidemiological study in South Korea to assess the prevalence of DICCD using the DSM-5 criteria. While international investigations estimated the prevalence of “any disruptive disorder” to be approximately 5.7% [7], such estimates typically lacked explicit operational definitions and predated the DSM-5 reorganization, limiting direct comparability. Given that DICCD in this sample was predominantly composed of ODD cases, the clinical interpretation is focused on ODD and CD as separate diagnostic entities.

The prevalence of ODD was 2.52%. According to data from previous international investigations, the prevalence of ODD was estimated to be 3.6% [6 , 7]. The assessment methodologies and diagnostic criteria are the primary sources of this difference. The current study employed the KSADS-COMP with the DSM-5 criteria, which requires comprehensive clinical interviews and more stringent functional impairment requirements than the DSM-III/IV-based instruments used in international studies. The DSM-5 introduced enhanced diagnostic thresholds, including specific duration requirements and severity specifiers, which effectively excluded borderline cases previously captured in earlier DSM versions. In contrast, the prevalence rate of ODD reported in a previous Korean community-based study was much higher, ranging from 3.7% to 4.9% [5 , 32]. The differences in sample representativeness were particularly relevant. Although our data were obtained from a nationally representative sample that included the general population of Korean children and adolescents, previous studies have focused on urban or suburban populations. These methodological differences collectively account for the lower prevalence rates observed in this study.

The prevalence of CD was 0.61% in this study. According to data from previous international investigations, the prevalence rate of CD is estimated to range from 1.9% to 2.1% [6 , 7]. Similar to ODD, the lower prevalence can be attributed to the more stringent DSM-5 diagnostic criteria and comprehensive clinical assessment methodology employed in the current study. In contrast, the prevalence rate of CD was 0.4%, as reported in a previous Korean community-based study [32], which is lower than that in the current study. This variation is primarily explained by differences in sample characteristics. Park’s sample was restricted to elementary school students aged 6–12 years from six urban school districts in Seoul [32], whereas the current study included 6- to 17-year-olds nationally. Given that CD prevalence increases substantially with age, peaking during adolescence, the inclusion of adolescents (12–17 years) in the current study naturally elevated the overall prevalence rate, despite the more conservative diagnostic approach.

This study identified a complex profile of risk factors for DICCD, ODD, and CD among Korean children and adolescents, providing key insights into behavioral problems among the youth. The DICCD risk factors included male sex, aged 12–17 years, employed mothers, parental depression and anxiety, child’s sleep problems, maternal alcohol use, and excessive screen use.

Common risk factors for both ODD and CD included age 15–17 years, parental anxiety problems, child sleep problems, and moderate game use (1–3 hours). Both disorders peak in early-to-mid-adolescence, which is consistent with prior research indicating an elevated risk of

behavioral disorders during adolescence [33]. Parental anxiety was confirmed as a shared risk factor for ODD and CD, consistent with prior studies [8 , 10 , 11 , 34 , 35]. on reciprocal parent-child well-being.

ODD and CD exhibited distinct risk factor profiles across multiple domains. Regarding demographic factors, male sex was significantly associated with CD (p=0.024), but only had a borderline association with ODD, which is consistent with a prior study that identified male sex specifically with CD [13]. Additionally, CD demonstrated broader developmental involvement across all age groups (9–17 years) compared to ODD's concentration in older adolescence (15–17 years), and rural residence emerged as a unique risk factor for CD. In the parental and family contexts, ODD was associated with higher maternal employment, parental depression, and maternal drinking (to 2–4 times monthly). Maternal employment, which has rarely been reported in international research, may reflect reduced direct caregiving time and opportunities for emotional feedback. Notably, although a meta-analysis found that maternal employment increases conduct problems [36], this association was not observed in the current study. In the screen use domain, ODD was associated with extensive screen exposure (video ≥3 hours daily, gaming ≥1 hour daily, educational content ≥3 hours daily), consistent with a prospective cohort study, whereas CD showed no significant association with social media use despite previous findings identifying it as a strong CD predictor [37]. These divergent risk patterns across demographic, family, and environmental domains suggest that the associations of risk factors differ between ODD and CD, supporting the development of disorder-specific prevention and intervention strategies.

The comorbidity risk patterns observed for DICCD, ODD, and CD in this study partially align with those of previous studies, while also revealing significant differences. DICCD, a relatively recent DSM-5 diagnostic category, has limited international systematic comorbidity research. Notably, this is the first comprehensive analysis of DICCD's comorbidity profile in the Korean population.

In our findings, DICCD demonstrated the highest comorbidity risk with mood disorders, followed by substance use and neurodevelopmental disorders. This indicates that DICCD is associated with a broad range of psychiatric conditions beyond behavioral dysregulation, underscoring the need for a comprehensive assessment at the time of diagnosis.

For ODD, prior research has reported high rates of psychiatric comorbidities in patients with ODD [18]. In our study, ODD exhibited the highest comorbidity risk with mood disorders, in contrast to a previous Korean elementary school study that found its strongest association with anxiety disorders [17]. This discrepancy likely reflects age differences; the previous Korean study examined younger children (mean age 9.22±1.48 years), while our sample included both children and adolescents. This interpretation is supported by international evidence; a systematic review of community and epidemiologic studies showed that major depression exhibited higher odds ratios with ODD than anxiety disorders, with sample characteristics such as age significantly influencing these estimates [38]. These findings highlight the necessity for systematic screening for mood disorders and the adoption of treatment approaches that address emotional difficulties in ODD. Moreover, the elevated comorbidity risk of neurodevelopmental disorders in patients with ODD observed in this study is consistent with international research documenting a significant association between ODD and neurodevelopmental disorders [39].

In CD, SUDs demonstrated the highest comorbidity risk, substantially exceeding that of other psychiatric conditions, which is consistent with previous reports of elevated SUD prevalence among adolescents with CD [40]. Of particular note is the high comorbidity risk observed for suicidal behaviors, suicidal ideation, and NSSI in CD, mirroring international studies that have documented

strong associations between CD and suicidal behavior [41 , 42], and research indicating higher rates of suicidal ideation in CD than in ODD [43]. Therefore, CD diagnosis should be accompanied by a systematic assessment of substance use and suicide risk, establishment of safety plans, and implementation of long-term suicide prevention interventions. Analyzing ODD and CD as distinct entities, rather than as a combined disruptive behavior category, has important clinical and public health implications. In this nationally representative sample, ODD and CD differed not only in prevalence but also in their risk factor and comorbidity profiles—for example, ODD showed stronger associations with mood disorders and psychosocial stressors, whereas CD was more strongly associated with SUDs and suicidal behaviors. These diagnosis-specific profiles suggest that prevention, screening, and intervention strategies should be tailored to the predominant risk and comorbidity patterns of each disorder, rather than applying a one-size-fits-all approach to disruptive behavior problems.

The strengths of the present study include the following: Unlike previous Korean studies that were geographically limited or analyzed ODD and CD in combination, this study is the first to use diagnosis-specific epidemiological data in Korean children and adolescents to measure the prevalence, risk factors, and comorbidities of disruptive, impulse control, and CDs using the KSADS-COMP diagnostic interview utilizing a nationally representative sample. This approach aligns with the international trend of analyzing ODD and CD as separate diagnostic categories and clearly demonstrates clinically important diagnosis-specific differences. Second, semi-structured clinical interviews using the KSADS-COMP enable precise diagnoses per DSM-5 criteria and provide substantially higher diagnostic accuracy than screening instruments or rating scales used in previous studies.

This study has some limitations. The DICCD in this sample was predominantly composed of ODD cases (n=158 ODD vs. n=38 CD among the total N=173 DICCD); accordingly, the DICCD-specific findings should be interpreted primarily as reflecting ODD characteristics rather than representing a distinct unified clinical entity. Therefore, subsequent clinical interpretation focuses on ODD and CD as separate diagnostic entities, consistent with their distinct clinical presentations and comorbidity profiles. Second, the cross-sectional design fundamentally limits causal inference. Temporal ordering and causal directions cannot be established, and reverse causality (e.g., child disruptive behaviors exacerbating parental distress or increasing screen use) as well as bidirectional processes are plausible; therefore, the observed relationships should be interpreted as associations rather than causal effects. Third, several lifestyle measures have inherent measurement limitations. Sleep problems were defined using uniform cutoffs across ages 6–17 years (e.g., total sleep duration <5 or >10 hours), which may not fully capture developmentally normative sleep variability, particularly in younger children. Although this definition was consistent with previous large-scale surveys, the binary classification may oversimplify the dimensional nature of sleep and should be interpreted with caution. Screen time was assessed by self- or caregiver-report and is therefore subject to recall bias and social desirability effects; combined with the cross-sectional design, the associations between screen use and DICCD/ODD should be regarded as exploratory. Fourth, we did not apply formal corrections for multiple testing despite conducting multiple logistic regression analyses across disorders and outcomes; although our analyses were hypothesisdriven and stringent corrections in rare conditions (e.g., CD, 0.61%) may obscure clinically meaningful associations, the risk of Type I error cannot be excluded, and findings with p values approaching the 0.05 threshold should be interpreted cautiously. Moreover, although sex, age group, maternal employment, parental depression, and parental anxiety were treated as core a priori confounders based on prior epidemiological and clinical evidence, our covariate selection did not rely on a formal causal modeling

framework (e.g., directed acyclic graphs), and residual confounding from unmeasured or imperfectly measured factors cannot be ruled out. Finally, the small sample size of the CD group (n=38) represents a substantial limitation that reduces statistical power, widens confidence intervals, and may compromise the stability of multivariable regression models. This limited sample size precludes detailed subgroup analyses and may fail to detect small effects or rare associations, requiring particularly cautious interpretation of CD-specific findings.

This study provides the first nationally representative epidemiological data on DICCD in Korean children and adolescents, revealing prevalence rates of 2.76% for DICCD, 2.52% for ODD, and 0.61% for CD. ODD and CD revealed distinct risk factors and comorbidity profiles: ODD was associated with psychosocial stressors, presenting with mood disorder comorbidity (OR 37.43), while CD is characterized by demographic-environmental factors and presents with crisis-level comorbidities, SUD (OR 60.49), and suicidal behavior (OR 16.93). These findings underscore the need for disorder-specific prevention strategies, comprehensive psychiatric screening protocols, and integrated treatment approaches targeting highrisk adolescents to address comorbid psychiatric conditions.